Reader Report: Week 05

Bayesian Invariant Prediction & Optimization-Based ICP

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Summary

Wu and Blei (2025) introduce **Bayesian Invariant Prediction** (BIP), casting ICP as latent-variable inference. A binary selector $z \in \{0,1\}^p$ indexes invariant features; the posterior $p(z \mid \mathcal{D})$ is proportional to likelihood ratios between pooled and per-environment conditionals. They prove consistency (Thm 1) and exponential contraction at rate $O(R \cdot e^{-\kappa n E \mu_{\min}})$ (Thm 2), where $\mu_{\min}$ captures environment heterogeneity. For scalability, **VI-BIP** uses mean-field variational inference with the U2G gradient estimator; it identifies invariant genes in a 6,170-gene perturbation dataset where competing methods falter.

Fan et al. (2024) propose **NegDRO**, a continuous minimax formulation

$$\min_b \max_{w \in U(\gamma)} \sum_e w_e \, \mathbb{E}[Y^e - b^\top X^e]^2,$$

where $U(\gamma)$ allows *negative weights* to enforce risk invariance. Under additive interventions, every stationary point is $O((1+\gamma|E|)^{-1} + T^{-1/4} + n^{-1/4})$-close to β^* (Thm 5), enabling gradient descent in *polynomial* time rather than 2^p enumeration. In short, BIP gives full posterior uncertainty over invariant sets, while NegDRO trades distributional characterization for polynomial-time convergence with provable rates.

Critique & Project Connection

My *Predicting NIMBYism Causally* project observes roughly four temporal environments and approximately fifty features. In this low-environment regime, BIP's multi-modal posterior honestly quantifies which invariant sets remain ambiguous rather than returning a single point estimate that may be an artifact of limited data. NegDRO's polynomial-time guarantee and tolerance of hidden confounders better suit iterative model development, since unmeasured wealth may simultaneously confound school quality and zoning opposition. A productive strategy is to run NegDRO first for fast screening, then apply BIP to the reduced feature set; features flagged as causal by both methods warrant confidence, while disagreements signal that additional environments are needed. Both methods assume linear models with squared loss, however, and our protest indicator $Y \in \{0,1\}$ requires either a link function or a loss-function generalization that preserves the invariance guarantee.

Questions

1. NegDRO identifies β^* when Condition 1b holds, requiring that second-moment matrices exhibit sufficient heterogeneity across environments ($\lambda > 0$). BIP's posterior contracts when $\mu_{\min} > 0$, a KL-divergence measure between local and pooled conditionals. These are distinct notions of "sufficient heterogeneity." Can environments satisfy one but not the other, and if so, which method degrades more gracefully when its specific heterogeneity condition is only weakly satisfied?
2. Both methods assume squared loss. For binary Y, the invariance guarantee therefore rests on a misspecified loss function. Does misspecification affect the two methods symmetrically? NegDRO enforces risk parity across environments, which may remain well-posed even under misspecification since equal risk is a relative condition. BIP's posterior, by contrast, depends on absolute likelihood evaluations, raising the possibility that it concentrates on the wrong z^* when the assumed Gaussian conditionals are far from the true Bernoulli data-generating process.

Project Update: Predicting NIMBYism Causally

1. What is the problem?

We seek the causal drivers of zoning opposition in Austin, TX. Following Peters et al. (2016), we adopt invariant causal prediction: find S^* such that $P_e(Y \mid X_{S^*}) = P(Y \mid X_{S^*})$ across all environments e. Features stable across regimes are causal; those whose predictive power breaks are spurious.

The key to identification is environment diversity: each regime change stress-tests predictors. Over 2019–2024, Austin experienced the mayoral transition (Jan 2023), parking-reform elimination (Nov 2023), the HOME Initiative (Feb 2024), and council-seat turnovers. These yield temporal and geographic environments (Bühlmann, 2020, assumption B), constituting **limited additive interventions** (Fan et al., 2024; Rothenhäusler et al., 2019). Peters et al.'s estimated set $\hat{S}(\mathcal{E}) \supseteq S^*$ is an outer bound that shrinks monotonically with each additional environment (Peters et al., 2016, Thm 1); Fan et al.'s Condition 4 extends identification to partial interventions, and BIP's posterior quantifies remaining ambiguity. With approximately fifty features, classical ICP's 2^p enumeration is infeasible, motivating both methods.

2. Mathematical Formulation

Let $\mathcal{E} = \{e_1, \ldots, e_K\}$ index the K regimes above. In each e we observe $(X_i^e, Y_i^e)_{i=1}^{n_e}$ with $X^e \in \mathbb{R}^p$ and $Y^e \in \{0,1\}$ (protest signed). Covariates partition into intervened S_L (regulatory/market features) and untouched S_L^c (demographics, schools, transit).

This partition motivates the linear SEM (Peters et al., 2017, Ch. 4) under the additive intervention regime (Rothenhäusler et al., 2019):

$$Y^e = (\beta^*)^\top X_{S^*}^e + \varepsilon_Y^e, \quad \varepsilon_X^e \stackrel{d}{=} \eta_X + (0_{S_L^c},\, \delta_{S_L}^e)^\top, \quad \mathbb{E}[\eta\,(\delta^e)^\top] = 0.$$

Here $S^* \subseteq [p]$ indexes the true causal parents; S_L the intervened covariates; S_L^c the untouched ones. Only $\delta_{S_L}^e$ varies, so $Y \mid X_{S^*}$ is stable and prediction risk $\mathbb{E}[(Y^e - (\beta^*)^\top X^e)^2] = \sigma_Y^2$ holds in every environment. This is the *risk invariance* we seek to exploit.

With roughly four environments and fifty features the invariant set is likely underdetermined, so we pair two methods that trade off differently. Fan et al. (2024) cast recovery as minimax,

$$\hat{\beta}^\gamma \in \arg\min_b \max_{w \in U(\gamma)} \textstyle\sum_e w_e\, \hat{\mathbb{E}}[(Y^e - b^\top X^e)^2],$$

where $U(\gamma)$ permits *negative* weights; under Condition 4 this yields a fast point estimate of β^* even with limited interventions. Wu and Blei (2025) place a prior over selectors $z \in \{0,1\}^p$ whose posterior

$$p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z) / \hat{p}_e(Y_i^e \mid X_{ei}^z)$$

concentrates on z^* but crucially reveals *which features remain ambiguous* when environments are few. Where the two agree we make confident causal claims; where they diverge the data cannot resolve S^*.

References

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